# CRYPTOCURRENCY STATISTICAL ARBITRAGE

### Momentum and reversal | Hourly, dollar-neutral research | 2022-2025

Course project report

## Research question

Can cross-sectional price patterns in eight cryptocurrencies predict subsequent returns strongly enough to survive 20 basis points of one-way trading costs? The project compares two predeclared strategy families and tests a volume-activity diagnostic.

## Principal finding

None of the frozen selections was profitable after the prescribed 20 bps cost in the 2025 test.

Mom 168h/24h: -40.38% net test return, Sharpe -2.75, maximum drawdown 42.60%.

Rev 1h/24h: -57.72% net test return, Sharpe -4.43, maximum drawdown 59.02%.

| Evidence | Design |
|---|---|
| Universe | BTC, ETH, LTC, ADA, LINK, DOGE, DOT, SOL; Binance USDT spot |
| Development / validation / test | 2022-2023 / 2024 / 2025; December 2021 warm-up |
| Experiments | 18 fixed combinations: 9 momentum and 9 reversal |
| Position model | Theoretical 50% long / 50% short; unit gross exposure |
| Execution | Close-based idealization; 0, 20, 40 bps and one-extra-hour delay |

## Interpretation boundary

This is an unconstrained educational backtest, not an implementable spot shorting system. Borrowing, funding, leverage constraints, execution capacity, and weight drift are excluded. The fixed survivor universe and multiple comparisons limit generalization. One documented 2023 exchange halt is handled as an observed four-hour holding interval, without fabricated prices.

Source of the assignment: ClassProject.docx, The Wall Street Quants course. Data attribution: Binance public market data. No claim of profitability was assumed in constructing this report.

Binance Vision data-derived analysis | CC BY-NC-SA 4.0 | Research only

# 02 / Research story and choice of universe

### Start with the course question

The assignment asks whether momentum or reversal in cryptocurrency price-volume data can survive trading costs. It permits a flexible research design but does not prescribe any coin names, a universe size, or an asset-selection rule. The original project operationalized that question with an hourly, cross-sectional backtest and a fixed universe before comparing strategy parameters.

### How the eight coins were defined

The configured instruments are BTCUSDT, ETHUSDT, LTCUSDT, ADAUSDT, LINKUSDT, DOGEUSDT, DOTUSDT and SOLUSDT. The retained report describes them as eight present-day recognizable coins and a **purposive survivor sample**. They were specified directly in the configuration and downloader; they were not produced by a recorded market-cap, liquidity, correlation or profitability screen. The precise reason for choosing each coin, or choosing eight rather than another number, is not documented.

Using a single exchange and the same USDT quote currency gives a consistent source and numeraire. The eight pairs have archived hourly data from December 2021 through December 2025, subject to the disclosed halt and metadata exceptions. These are practical advantages of the setup, not evidence that they were formal historical selection criteria.

### What was the bigger pool?

**No larger candidate pool was enumerated in the retained project.** There is no recorded initial coin count, rejected-asset list, ranking, or point-in-time eligibility table. All Binance spot USDT pairs would be a natural broader opportunity set for a different study, but the original project did not demonstrate that it screened that set. It would therefore be incorrect to claim that these were the eight most liquid coins or that only eight survived a market-wide screen.

### Data checking is not universe selection

Acquisition started with the eight configured pairs and verified 392 monthly archives: 49 per pair. The subsequent audit checked coverage, checksums, duplicates, OHLCV consistency and the documented exceptions. It did not compare these eight with excluded coins. The 18 candidates in the original research were **strategy configurations** (nine momentum and nine reversal), not 18 competing coins; 2024 selected signal parameters while the asset universe stayed fixed.

### What this means for the story

The findings are conditional on this fixed survivor universe, not a point-in-time representation of the full crypto market. Failed, delisted and newly listed alternatives were not represented, so generalization is limited. This revised report preserves the original 18-configuration study and its initial 2025 evaluation; later exploratory scans and the 2026 historical holdout remain separate.

**Universe evidence:** supplied ClassProject.docx; configs/research.toml; src/crypto_stat_arb/data.py; data/raw/manifest.json; data/processed/data_quality.json; README.md; and the preserved report implementation. No retrospective asset-screening story has been invented.

# 03 / Data and experimental protocol

## Input and provenance

Hourly spot OHLCV comes from Binance public monthly archives. Original archives and SHA-256 checksums are cached; normalized Parquet inputs retain completed UTC-hour endpoints. The loader accounts for the 2025 spot timestamp-unit change. All eight instruments use the same USDT quote currency.

| Split | Period | Permitted use |
|---|---|---|
| Warm-up | December 2021 | Trailing indicators only |
| Development | 2022-2023 | Implementation and diagnostic design |
| Validation | 2024 | Select one configuration per family |
| Test | 2025 | Frozen-selection evaluation; no retuning |

## Predeclared grid

| Family | Signal lookback | Rebalance spacing |
|---|---|---|
| Momentum | 24, 72, 168 hours | 1, 4, 24 hours |
| Reversal | 1, 4, 12 hours | 1, 4, 24 hours |

## Quality gate

Checks cover duplicates, missing hours, OHLC consistency, volumes, checksums, and cross-instrument alignment. Status is pass with documented exceptions, not a clean complete-hourly pass. All missing prices remain missing; the full audit is saved with the dataset.

| Observed quality field | Recorded value |
|---|---|
| Status | pass_with_documented_exceptions |
| Rows | 286456 |
| Archive Count | 392 |
| Missing Count | 8 |
| Duplicate Count | 0 |

## Predeclared exception: March 24, 2023 spot halt

Binance reports a halt at 11:27 UTC and resumption at 14:00. The missing 14:00 endpoint affects all eight assets and is not repaired by official daily archives. We bridge observed closes 11:00 to 15:00 as one four-hour interval, hold the existing position, and book the complete price-ratio return once at 15:00. No executions, costs, valuations, price filling, early cash-out, or dropped day occur at 12:00-14:00. Official halt account.

Sixteen close-time metadata anomalies are documented: one SOL warm-up record is corrected from a checksum-verified daily archive without changing OHLCV; fifteen shortened close timestamps are allowlisted warnings. No observed prices are changed. Unknown exceptions still stop research.

## Selection rule

Maximize 2024 net Sharpe per family; tie-break by lower turnover, then identifier. A 50/50 position blend requires both families to pass positive net validation. Blend costs come from combined positions.

# 04 / Signals, portfolio, and accounting

## Signal construction

For coin i at completed hour t, trailing return is close(i,t) / close(i,t-L) - 1. Momentum uses this return; reversal uses its negative. Cross-sectional average ranks are demeaned, then normalized by the sum of absolute demeaned ranks. Ties receive average ranks; an all-tied cross-section is flat.

| Operation | Definition |
|---|---|
| Cross-sectional score | z(i,t) = rank(signal(i,t)) - mean(rank(signal(t))) |
| Target weights | w(i,t) = z(i,t) / sum_i abs(z(i,t)) |
| Gross next-hour return | r_gross(t+1) = sum_i w(i,t) * r(i,t+1) |
| Traded notional | turnover(t) = sum_i abs(w_new(i,t) - w_old(i,t)) |
| Cost and net return | cost(t) = turnover(t) * bps / 10,000; r_net = r_gross - cost |
| Compounded equity | equity(t) = product over s<=t of [1 + r_net(s)] |
| Drawdown | equity(t) / max(1, earlier equity values) - 1 |

## Time alignment and costs

Targets are set only on UTC-aligned rebalance hours and held between decisions, except no execution is allowed during the documented halt. Completed-bar signals earn subsequent, not contemporaneous, returns. Each split starts from cash, charges initial entry, and includes final liquidation. Halt lookbacks require exact elapsed-hour endpoints: unavailable signals defer the decision while retaining exposure. A long-to-short change pays the full absolute weight change. The extra-delay test moves signals back one additional hour.

This close-based execution assumption is deliberately idealized: the observed close is not a guaranteed executable price. Twenty bps is the course-prescribed all-in market-order cost, not an empirical estimate from an order book. Seven-bps limit-order commissions do not justify assuming fills without spread, adverse selection, or opportunity costs.

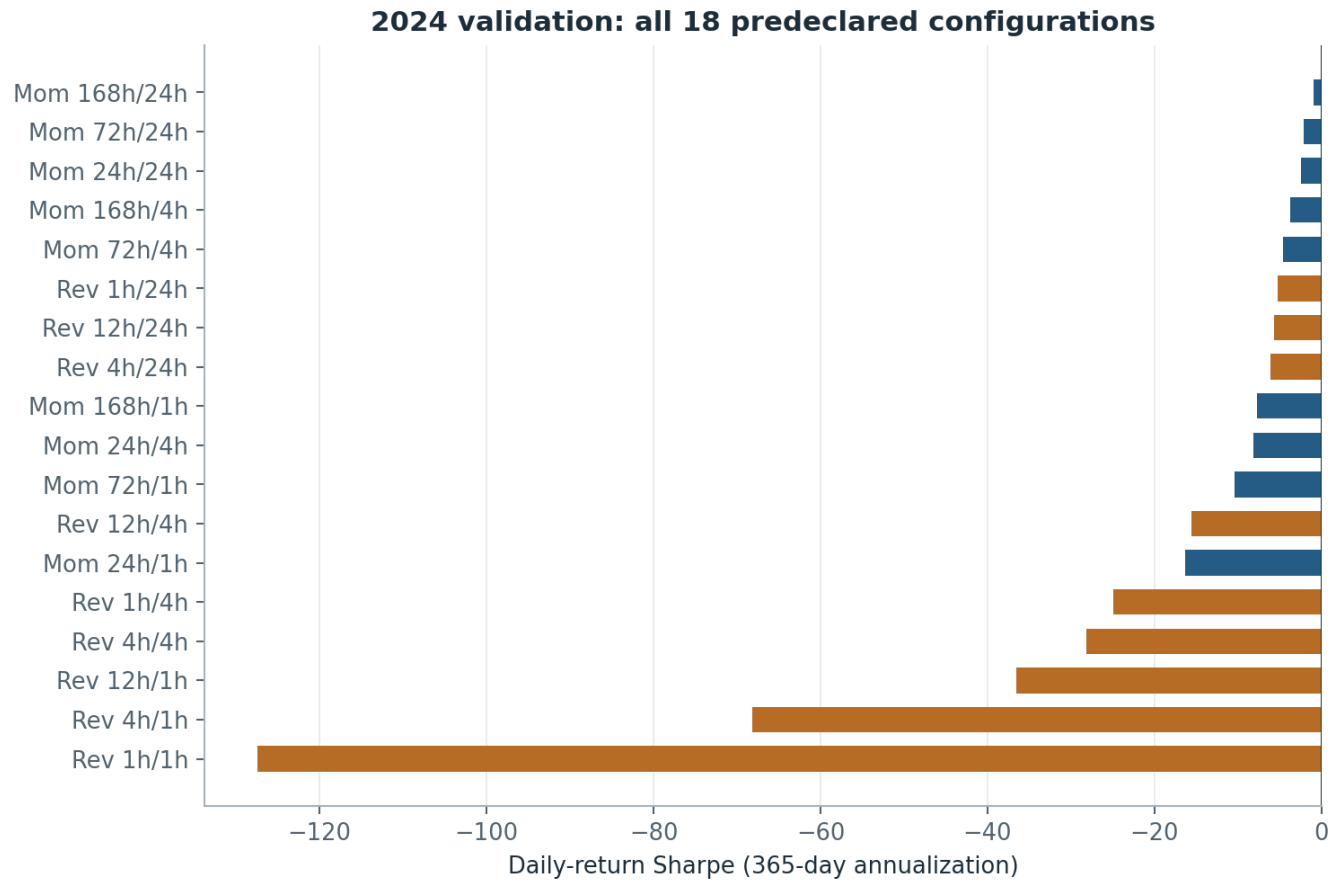
## Performance definitions

Observed-interval net returns compound into UTC daily returns with exactly 24 elapsed hours of coverage; the 2023 bridge preserves the reopening return. Arithmetic annual return = daily mean x 365; annual volatility = daily sample standard deviation x sqrt(365); Sharpe uses a zero risk-free rate. CAGR uses compounded equity and elapsed years; maximum drawdown uses observed-endpoint equity including initial capital. It is hourly outside the halt; intrahalt losses cannot be reconstructed.

Daily OLS against BTC/USDT returns includes an intercept. Reported alpha is annualized by 365 and beta is dimensionless. Alpha uncertainty uses HAC standard errors with seven daily lags; this handles a specified short autocorrelation window but does not correct configuration-search bias. USDT is the numeraire, not guaranteed identical to USD.

Course-style approximation: target changes determine turnover. We do not track weight drift, cash financing, margin, mark-to-market collateral, or capacity.

# 05 / Validation and frozen selections




The comparison uses the same 2024 period and prescribed execution costs for all 18 configurations. Color distinguishes the two families; the zero line separates positive and negative Sharpe. Labels show family and lookback/rebalance hours. No test-period metric is used to choose winners.

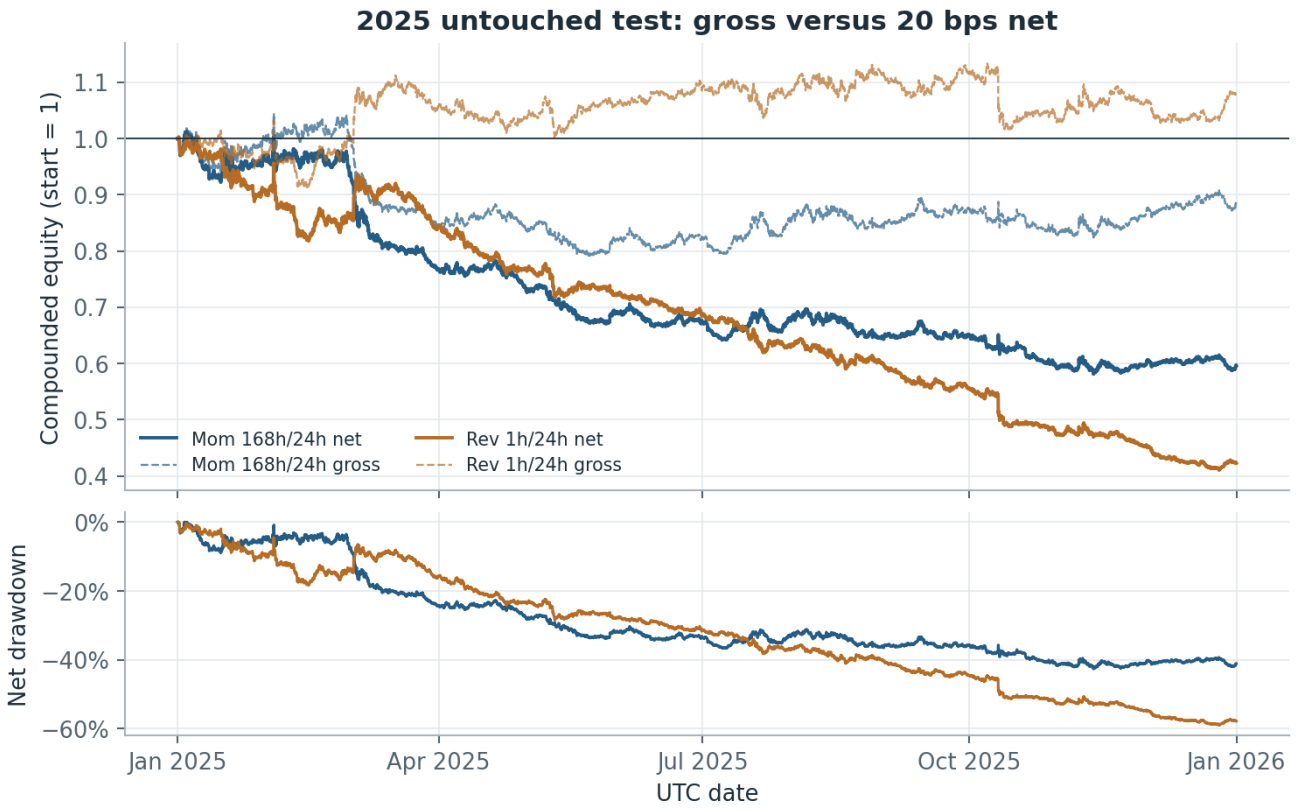
| Frozen strategy | Net return | Sharpe | Max drawdown |
|---|---|---|---|
| Mom 168h/24h | -22.96% | -0.98 | 41.92% |
| Rev 1h/24h | -71.67% | -5.20 | 73.22% |

## Why validation is not proof

Selecting the best of nine candidates within a family can capitalize on noise. All configurations are retained in the experiment table, and evaluation advances chronologically into 2025. The held-out year reduces selection leakage, but one market regime and a survivor-selected sample cannot establish a persistent trading edge.

Selection provenance is saved separately in outputs/selection.json. The runner writes that frozen decision before creating test metrics. Execution-cost and delay sensitivity cases reuse those decisions; they do not trigger reselection.

# 06 / Untouched 2025 test results

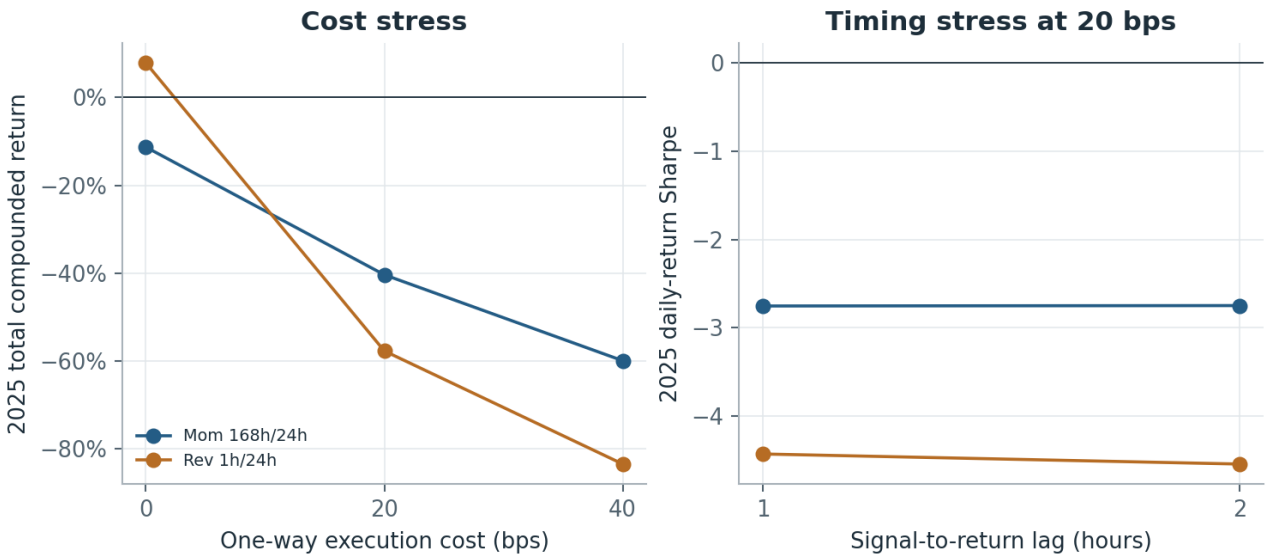



| Metric | Mom 168h/24h | Rev 1h/24h |
|---|---|---|
| Total return | -40.38% | -57.72% |
| CAGR | -40.38% | -57.72% |
| Arithmetic annual return | -50.03% | -84.17% |
| Annual volatility | 18.17% | 19.00% |
| Sharpe (zero risk-free) | -2.75 | -4.43 |
| Max hourly drawdown | 42.60% | 59.02% |
| Annual BTC-relative alpha | -49.85% | -84.18% |
| BTC beta | -0.08 | 0.00 |
| Alpha 95% HAC interval | -86.92% to -12.79% | -119.34% to -49.02% |

None of the frozen selections was profitable after the prescribed 20 bps cost in the 2025 test.

Solid curves are 20 bps net equity; dashed curves are gross before transaction costs. Net drawdowns below use the same start-at-one capital base. The table is net at the minimum declared lag; alpha and its 95% HAC interval are expressed in annual return units; positive maximum drawdown is loss magnitude. N/A means unavailable, never zero.

# 07 / Execution sensitivity and robustness




| Strategy | Cost bps | Lag h | Net return | Sharpe |
|---|---|---|---|---|
| Mom 168h/24h | 0 | 1 | -11.32% | -0.57 |
| Mom 168h/24h | 20 | 1 | -40.38% | -2.75 |
| Mom 168h/24h | 40 | 1 | -59.94% | -4.89 |
| Mom 168h/24h | 20 | 2 | -40.13% | -2.75 |
| Rev 1h/24h | 0 | 1 | 7.85% | 0.49 |
| Rev 1h/24h | 20 | 1 | -57.72% | -4.43 |
| Rev 1h/24h | 40 | 1 | -83.46% | -9.29 |
| Rev 1h/24h | 20 | 2 | -59.22% | -4.54 |

| 20 bps baseline | Gross return | Net return | Mean hourly turnover |
|---|---|---|---|
| Mom 168h/24h | -11.32% | -40.38% | 0.023 |
| Rev 1h/24h | 7.85% | -57.72% | 0.053 |

Turnover is traded notional per unit of target capital, not half-turnover. Mean hourly turnover includes entry and liquidation; gross-net equity differences are compounded cost drag, not summed hourly cost rates.

### Reading the stress cases

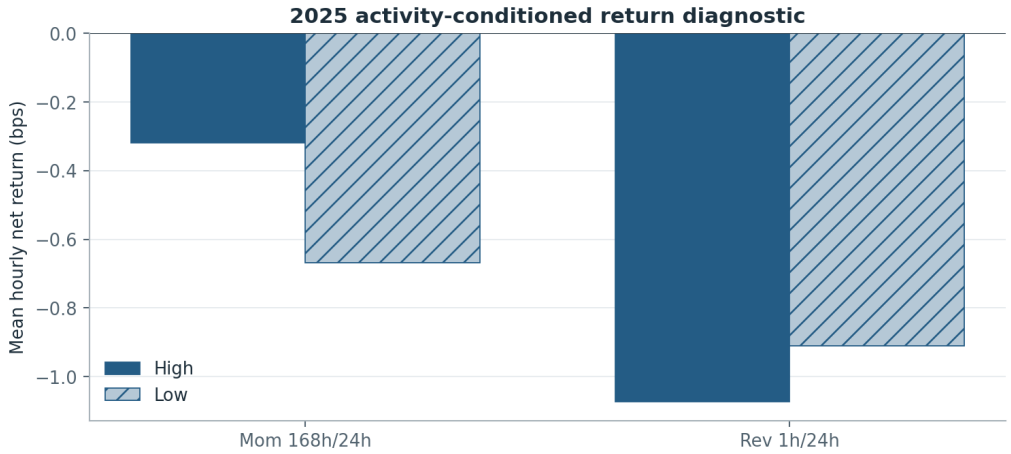
Zero cost measures gross signal behavior, not achievable profit. Higher costs test implementation sensitivity; an extra hour tests close-timing dependence. These are stresses, not confidence intervals; no 2025 reselection occurs.

Even a positive scenario is insufficient for deployment. A subsequent implementation study would need borrowable instruments, funding and financing, measured spreads, order-book capacity, realistic execution, and an additional untouched period.

# 08 / Volume activity and time stability

## Activity as a descriptive proxy

For each coin, current volume is divided by its preceding 168-hour mean, excluding the current bar from that denominator. The marketwide activity score averages these relative volumes. The median estimated only on development data divides high and low activity. Missing preceding-volume coverage is an explicit unknown regime, not fabricated low activity. The activity label follows the strategy rebalance decision and execution lag, excluding later activity information; its use is descriptive, not an additional trading gate.




| Strategy | Regime | Hours | Mean net (bps/hour) |
|---|---|---|---|
| Mom 168h/24h | high | 2544 | -0.319 |
| Mom 168h/24h | low | 6216 | -0.667 |
| Rev 1h/24h | high | 2544 | -1.074 |
| Rev 1h/24h | low | 6216 | -0.909 |

Higher trading volume may reflect information, liquidity demand, volatility, or mechanical trading. This split cannot identify informed versus uninformed traders or establish a causal mechanism. Means across discontinuous subsets are not annualized Sharpes; counts matter and serial dependence remains. Zero-hour unknown regimes are retained in the CSV, not displayed above.

## Year-by-year stability

| Strategy | Year | Net return | Sharpe |
|---|---|---|---|
| Mom 168h/24h | 2022 | 12.07% | 0.59 |
| Mom 168h/24h | 2023 | -16.14% | -0.72 |
| Mom 168h/24h | 2024 | -22.96% | -0.98 |
| Mom 168h/24h | 2025 | -40.38% | -2.75 |
| Rev 1h/24h | 2022 | -52.06% | -2.95 |
| Rev 1h/24h | 2023 | -65.01% | -5.29 |
| Rev 1h/24h | 2024 | -71.67% | -5.20 |
| Rev 1h/24h | 2025 | -57.72% | -4.43 |

Annual results reset cash and liquidate each year. Earlier-year results for validation-selected configurations are retrospective; they are not independent yearly selections.

# 09 / Conclusions, limitations, and reproduction

## Answer to the course research question

None of the frozen selections was profitable after the prescribed 20 bps cost in the 2025 test.

Momentum and reversal were both implemented and compared under a common unconstrained portfolio rule. The decisive question is not whether some gross curve rises but whether the frozen signal remains economically meaningful after turnover, costs, and timing sensitivity. Results above answer that question only for this dataset and model.

## Material limitations

**Theoretical short side.** Spot price series supply returns, not borrowing availability. Funding, margin, financing, collateral, and liquidation risk are excluded.

**Survivor selection.** These eight present-day recognizable coins omit failed, delisted, and newly listed alternatives; this is not a point-in-time crypto-wide universe.

**Execution.** Fixed costs and next-return close alignment do not model spread variation, order-book impact, adverse selection, failed orders, or price latency.

**Statistical uncertainty.** Eighteen comparisons, one validation year, and one test year leave substantial false-discovery and regime risks. HAC alpha errors are not search-adjusted and cannot substitute for future validation.

**Data and numeraire.** Unknown outages or missing bars stop research; exchange-specific prices and USDT denomination limit portability. One four-hour 2023 holding interval is observed only at its endpoints; volume regimes become unknown when preceding volume coverage is missing.

**Approximate accounting.** Target-change turnover ignores portfolio weight drift. Daily annualization uses 365 days rather than equity-market calendars.

## Reproduction

The Python package, TOML configuration, checksum-backed cache, processed Parquet data, frozen selection manifest, and executed notebook are delivered together. The notebook loads local locked inputs, demonstrates the rank-neutral construction, independently rebuilds selected 2025 performance through the package, and reconciles saved returns and metrics. It does not download or retune strategies.

## Sources and attribution

1. **Course assignment:** The Wall Street Quants, *ClassProject.docx*; supplied locally. The report uses its unconstrained backtest framing and prescribed 20 bps market-order cost.

2. **Binance public market data:** archive schema, timestamp conventions, and checksum documentation. github.com/binance/binance-public-data.

3. **Dataset terms:** data-derived report, charts, and results are CC BY-NC-SA 4.0; non-commercial educational/research use with attribution. Binance data terms. No exchange endorsement is implied.

4. **Observed-halt protocol:** Binance March 24, 2023 halt explanation and official resumption announcement.